ECX2400 Homework 2

Empirical Paper Summary

Chosen Paper

Scaling up sanitation: Evidence from an RCT in Indonesia

Compared Paper

Krueger (1999), *Experimental Estimates of Education Production Functions*

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1 Summary of the Paper

1.1 Research question and Why it matters

The paper studies whether Community-Led Total Sanitation (CLTS) remained effective when scaled up in rural Indonesia. It asks whether CLTS increased toilet construction, reduced tolerance of open defecation, lowered child roundworm infection, and improved broader child health outcomes such as anemia, height, and weight. It also examines why the effects differed by poverty status, implementer type, and baseline social capital.

This question matters because sanitation creates health externalities. Open defecation can contaminate the shared environment, so one household's behaviour may harm the whole community. Since households may not fully internalise these social costs, private sanitation investment may be below the socially optimal level. CLTS tries to solve this collective-action problem through community mobilisation and social pressure rather than direct subsidies. Therefore, the paper is policy-relevant because it tests whether a low-cost and widely used sanitation program remains effective when scaled up.

1.2 Data

The paper uses primary data from a World Bank RCT of CLTS in rural East Java, Indonesia. The sample includes about 2,000 households across 160 villages in eight districts. The baseline survey was conducted in August–September 2008, and the endline survey was conducted about 24 months later, from November 2010 to February 2011. The data combine household surveys, physical toilet inspections, child fecal and blood samples, anthropometric measures, village information, social capital data, and implementation records. The unit of analysis is the household for sanitation outcomes and children aged 0–5 for health outcomes.

1.3 Empirical Strategy

The paper uses a village-level randomized controlled trial. Villages were randomly assigned to CLTS or control, so the control group shows what would likely have happened without the program. Because some villages did not follow their original assignment, the authors estimate an intention-to-treat effect, comparing villages by assignment rather than actual participation.

$$Y_{ij} = \alpha + \beta_1 T_j + \gamma' X_{ij} + \delta_k + \varepsilon_{ij}. \tag{1}$$

Here, Y_{ij} is the outcome for household or child i in village j, such as toilet construction, open-defecation attitudes, or child health. T_j equals one if village j was assigned to CLTS, so β_1 measures the causal effect of CLTS assignment under randomisation, not actual participation. X_{ij} includes baseline household and village controls, and δ_k controls for sub-district fixed effects from the stratified design. Standard errors are clustered by village because treatment was assigned at the village level. The authors also examine heterogeneous effects by poverty status, implementer type, and baseline social capital.

1.4 Main Findings

The paper finds that CLTS had positive but limited effects. It increased toilet construction by about 2.4 percentage points, a 19% increase relative to control villages, reduced tolerance of open defecation, and lowered child roundworm infection by around 46%. However, it did not significantly improve anemia, height, weight, or the overall child health index.

The effects were uneven. Poorer households were less able to improve sanitation because toilet construction required private investment and CLTS provided no subsidies. CLTS worked better when implemented by resource agencies, while local government implementation produced no clear benefits. Finally, the program was more effective in high-social-capital villages, but could be counterproductive in low-social-capital villages.

1.5 Table Analysis

Does CLTS treatment improve sanitation and child health outcomes?

Dependent Variables:	Toilet Construction		Intolerance of Open Defecation		Diarrhea Knowledge	Roundworm(eggs/g)		Hemoglobin(g/l)	Weight z-score	Height z-score	Health Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Treatment	0.024		0.409		0.007	−72.81		−0.18	0.03	−0.01	0.02
	(0.014)*		(0.233)*		(0.058)	(39.55)*		(0.51)	(.03)	(.03)	(.02)
RA Treatment* Poor		−0.027		0.644			−71.4				
		(0.034)		(0.56)			(110.5)				
RA Treatment* Non-poor		0.072		1.035			−176.4				
		(0.028)***		(0.312)***			(67.2)***				
LG Treatment* Poor		−0.028		0.069			35.9				
		(0.036)		(0.553)			(83.8)				
LG Treatment* Non-poor		0.014		−0.128			4.9				
		(0.016)		(0.316)			(48.4)				
Poor		0.009		−1.031			−69.9				
		(0.022)		(0.359)***			(65.9)				
Mean DV (Treat = 0)	0.125	0.125	32.9	32.9	4.85	156.7	159.2	111.0	−1.39	−1.65	0.001
Controls:	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Tests of Equality (p-values):											
RA* Non-Poor = RA* Poor		0.02		0.49			0.37				
LG* Non-Poor = LG* Poor		0.31		0.74			0.37				
RA* Non-Poor = LG* Non-Poor		0.06		0.004			0.01				
RA* Poor = LG* Poor		0.98		0.40			0.31				
RA* Poor = RA* Non-poor = LG* Poor = LG* Non-poor		0.09		0.04		0.68	0.09				

Original Table 2: Does CLTS treatment improve sanitation and child health outcomes?

Table 2 reports the paper's main intention-to-treat estimates for CLTS in rural East Java. It compares villages assigned to CLTS with control villages, using outcomes measured about 24 months after the August–September 2008 baseline survey. The unit of observation is the household for sanitation outcomes, such as toilet construction and attitudes toward open defecation, and children aged 0–5 for health outcomes. The table shows that CLTS had positive but limited effects: it increased toilet construction by 2.4 percentage points, reduced tolerance of open defecation, and substantially reduced child roundworm infection, but had no significant effect on hemoglobin, height, weight, or the overall health index. These results connect directly to the research question because they test whether CLTS improved sanitation behaviour and child health. They also matter for the paper's argument because the stronger effects are concentrated among non-poor households in villages implemented by resource agencies, while local government implementation produced little clear benefit. Overall, the table shows that CLTS partly worked, but its effects were modest, uneven, and incomplete when scaled up.

2 Connection to ECX2400 Course Content

2.1 Randomized Controlled Trials and Causal Effects (Lecture 2)

An RCT randomly assigns units to treatment and control groups, so the control group provides a credible counterfactual. This concept appeared in Lecture 2, where random assignment was used to identify causal effects. The paper applies it through a village-level RCT in rural East Java, comparing villages assigned to CLTS with control villages. The key assumption is that randomisation made the two groups comparable, which is plausible because the paper reports baseline balance and the assignment was stratified by sub-district. However, imperfect compliance means the estimate should be interpreted as an intention-to-treat effect: the effect of assignment to CLTS, not actual participation.

2.2 Heterogeneous Treatment Effects and Interaction Terms (Lecture 7)

Heterogeneous treatment effects mean the same program may affect different groups differently. This connects to Lecture 7, where interaction terms let an estimated effect vary across groups or conditions. The paper uses this idea by interacting treatment with poverty status, implementer identity, and baseline village social capital. This matters because the average treatment effect may hide unequal impacts: the paper finds stronger effects for non-poor households, villages treated by resource agencies, and villages with higher social capital. The key assumption is that these subgroup differences are not driven by other unobserved differences. This assumption is weaker than the main RCT assumption because poverty, implementer type, and social capital were not all randomly assigned. Therefore, these results are useful for policy interpretation, but should be interpreted more cautiously than the main ITT estimate.

3 Critical Reflection

One substantive limitation is that the paper does not directly test possible spatial spillovers across villages. The RCT treats control villages as a clean counterfactual, but sanitation behaviour may create externalities beyond village boundaries. For example, treated villages may reduce contamination in shared rivers or water sources, or spread sanitation information through nearby markets and social networks. If some control villages indirectly benefited from nearby treated villages, the treatment-control difference would be smaller than the true effect of CLTS. This matters because the paper concludes that CLTS had modest and incomplete effects; spillovers would mean the intention-to-treat estimates may understate the program's impact, especially for community-level health outcomes.

A feasible extension would be to collect GPS coordinates, river-network links, and information on shared water sources between villages. The authors could then construct a spillover-exposure measure, such as the number of treated villages within five kilometres, and add a robustness table controlling for nearby treatment exposure. They could also compare the main results with estimates using only control villages located far from treated villages. A map of treatment status and health outcomes would further show whether effects are spatially clustered.

Personal reflection. One point I found confusing was why CLTS reduced roundworm infection but did not improve broader health outcomes such as anemia, height, or weight. Thinking about possible spillovers helped me see that if some control villages were indirectly affected, the estimated treatment-control difference in health outcomes could be smaller than the true community-level effect.

4 Application and Comparison

I compare Cameron, Olivia, and Shah (2019) with Krueger's (1999) study of the Tennessee STAR experiment, discussed in Lecture 2 as a randomized controlled trial. Krueger examines how assignment to small classes, regular classes, or regular classes with a teacher aide affected student achievement, measured by standardised test scores.

Both papers use randomized controlled trials to estimate causal effects. In the CLTS paper, villages were assigned to treatment or control groups; in Krueger (1999), students and teachers were assigned to class-size groups within schools. In both cases, randomization is the identification strategy because treatment status should not be systematically related to other determinants of outcomes. Both papers also use regression to account for the experimental design: Cameron, Olivia, and Shah include sub-district fixed effects, controls, and village-clustered standard errors, while Krueger includes school fixed effects and student and teacher controls. Both papers examine heterogeneous effects: the CLTS paper focuses on poverty, implementer type, and social capital, while Krueger examines differences by student background.

The key difference is the trade-off between experimental control and real-world implementation. Krueger studies a clearer and more tightly controlled treatment: reducing class size, with achievement measured by standardised tests. This gives Krueger stronger internal validity. By contrast, CLTS is a complex community-level intervention that must change attitudes, sanitation investment, open-defecation behaviour, disease exposure, and child health. Its estimates are therefore more exposed to non-compliance, implementation differences, and possible spillovers across villages, so the main estimates are interpreted as intention-to-treat effects. However, this complexity is also why the CLTS paper is valuable: it tests whether a policy works under realistic scale-up conditions, not only in a clean experimental setting. Overall, Krueger is more convincing for isolating a causal mechanism, while Cameron, Olivia, and Shah are more informative for real-world policy design.